

Onatola Timilehin Faruq

Smart Contract Engineer | DeFi Protocol Integration | Security Tooling

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Summary

Solidity engineer with production experience integrating Aave v3, Uniswap v3/v4, Curve, and GMX. Built Sentinel, an automated smart contract auditing framework with Slither/Mythril integration, flash loan and oracle manipulation detection, and Foundry PoC generation. Strong Foundry testing practices (95%+ coverage), gas optimization (35% cost reduction), and security review track record (15+ contracts). Python proficiency, intermediate Rust, fully remote and async-first.

Technical Expertise

Smart Contracts: Solidity (Advanced) | Foundry (Forge, fuzz testing, fork testing, invariant testing) | Hardhat
DeFi Protocols: Aave v3 | Uniswap v3/v4 | Curve v1/v2 | GMX | Chainlink (Data Feeds, VRF, Automation, CCIP)
Security: Slither | Mythril | Semgrep | Flash Loan Analysis | Oracle Manipulation Detection | MEV Awareness | Audit Tooling
Languages: Solidity | Python | TypeScript | Rust (Intermediate, Anchor/Solana) | JavaScript
Monitoring & Analysis: Tenderly | Dune Analytics | The Graph (Subgraphs)
Infrastructure: GitHub Actions CI/CD | Docker | PostgreSQL | Redis
Chains: Ethereum | Base | Arbitrum | Optimism | BNB Chain | Solana | Hedera

Professional Experience

Blockchain Engineer | DeFiConnectCredit | Jan 2025 - Present

- Architected modular smart contract systems integrating **Aave v3, Uniswap v3/v4, Curve, and GMX** in production
- Established Foundry testing framework with **95%+ code coverage** including fuzz testing and fork testing against mainnet state
- Identified and resolved **3 critical vulnerabilities** pre-deployment through systematic security review (oracle manipulation, reentrancy)
- Reduced transaction costs by **35%** through assembly optimization and storage pattern improvements
- Built deployment automation suite with verification checks, cutting deployment time from 2 hours to 15 minutes

Blockchain Engineer (Intern) | DeFiConnectCredit | Jun 2024 - Dec 2024

- Conducted security reviews on **15+ contracts** pre-deployment, identifying reentrancy, front-running, and oracle manipulation vulnerabilities
- Developed test suites with **90%+ coverage** including edge case testing, invariant testing, and stress testing
- Contributed to collateral management and liquidation mechanism implementation for core lending protocol
- Built ERC721 incentive system with Merkle tree proofs for 500+ users

Security & Protocol Tooling

Sentinel — Automated Smart Contract Auditing Framework

Multi-agent security auditor for competitive audits (Sherlock, Code4rena, Cantina). Written in Python.

- Architected multi-agent pipeline: reconnaissance, static analysis (Slither/Mythril), LLM-powered deep analysis, PoC generation
- Built specialized vulnerability hunters for **flash loan attacks, oracle manipulation, reentrancy** (classic, cross-function, cross-contract, read-only), and access control
- Implemented automated **Foundry-based PoC generation** with mainnet fork testing and validation
- Supports 4 languages: Solidity, Rust/Solana, Move (Aptos/Sui), Cairo/StarkNet
- Integrated Semgrep rules for static pattern matching with cost-aware budget allocation

Python Slither Mythril Foundry Semgrep

DeFi Projects

Sentiment — Uniswap v4 Dynamic Fee Hook

UH17 Incubator Capstone | Protocol-level hook integrating on-chain and off-chain infrastructure

- Designed counter-cyclical fee algorithm (0.25%-0.44%) responding to real-time market conditions from 8 off-chain sources
- Built multi-keeper architecture using **Chainlink Automation + Gelato** for redundancy and failover
- Achieved **<30k gas overhead** per swap through optimization
- Graduated from Uniswap Hook Incubator (15% acceptance rate, Uniswap Labs collaboration)

Solidity Foundry Chainlink Uniswap v4

TruthBounty — Prediction Market Reputation Protocol

2nd Place, Seedify Hackathon | BNB Chain | Oracle-integrated reputation system across 13 markets

- Architected data verification pipeline across Polymarket, Azuro, Thales, PancakeSwap Prediction, and 9 others
- Designed soulbound NFT credentials with on-chain SVG generation
- Built custom subgraphs for cross-chain data indexing

Solidity The Graph BNB Chain Soulbound NFTs

Nocturne — Private Payments on Solana

Rust/Anchor with Noir/UltraHonk ZK proofs and TEE relay

- Built in **Rust/Anchor** with zero-knowledge proof circuits for withdrawal verification
- Implemented stealth addresses (DKSAP), Merkle commitment pools, TEE relay
- Achieved **97% unlinkability score** through systematic privacy testing

Rust Anchor Noir Zero-Knowledge

OmniSwap SDK — Cross-Chain Swap SDK

Production-grade TypeScript SDK with 70KB+ documentation

- Architected chain-specific adapter pattern across 6 blockchains with HTLC-based atomic swap protocol
- Built typed error system, exponential backoff retry logic, and automated refund management

TypeScript Multi-Chain SDK Design

Certifications & Training

- **Uniswap Hook Incubator (UHI7)** — Graduate (15% acceptance rate)
- **Cyfrin Updraft** — DeFi Security & Protocol Engineering (Aave v3, Uniswap, GMX, Curve, Auditing)
- **School of Solana** — Ackee Blockchain (Rust/Anchor Development)
- **Hashgraph Developer Certification** — The Hashgraph Association

Education

BSc Physiotherapy | University of Ibadan, College of Medicine | 2024

Portfolio: faruukku.vercel.app | GitHub: github.com/Leihyn | References available upon request